

Linux Programming: Assignment-7:

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1. What is a bash shell script? Give one example.

Ans:

Bash shell script is a text file containing series of linux commands

It is mainly used to automate repetitive task

A Bash script usually starts with a shebang (#!) line that tells the system which interpreter to use.

A Bash shell script is a set of commands saved in a file to automate tasks, making Linux system management faster and easier.

2. Write a simple shell script to print "Hello World".

Ans:

```
#!/bin/bash
```

```
echo "Hello World"
```

"echo" is used to print in bash script

3. What is the purpose of comments (#) in a shell script?

Ans:

The purpose of comments in shell script is to add the explanation within the code that are ignored by the shell during execution

Comments do not affect how the script runs

The first line in bash is a special case as it tells the system which shell to use

first line: `#!/bin/bash`

4. How do you declare variables (int, float, double, string, Boolean, and char in a shell script?)

Ans:

In bash variables are not declared using data types

All the variables are considered as string by default but they can hold numbers or text depending on the users use them

Syntax:

`variablename=value`

There should not be a space before or after "="

For int `num=10`

Float `pi=3.14`

String `name=jyothsna`

Boolean `flag=true`

Character `letter=A`

5. Write a shell script to display the current date and time of the system.

Ans:

```
#!/bin/bash
```

```
echo "Current Date and Time:"
```

```
date
```

Output:

Current Date and Time:

Sat Oct 11 12:45:00 IST 2025

6. Explain the difference between a constant and a variable in bash script.

Ans:

Variable: a variable is a value that changes during execution of a script. We can assign, modify or use this value anywhere in the script. Variables store the value you give in your script and you can modify it anytime you want.

Constant: a constant is a fixed value in the script that cannot be changed once it is defined. But users can create their own constants by using `readonly` or `declare -r` command. Constant values cannot be modified once they are assigned a value in script.

7. Write a shell script to read two integer numbers from the user and compute the sum of both numbers.

Ans:

```
#!/bin/bash
echo "Enter the first number:"
read n1
echo "Enter the second number:"
read n2
sum=$((n1 + n2))
echo "The sum of $n1 and $n2 is: $sum"
```

Output:

Enter the first number:

10

Enter the second number:

20

The sum of 10 and 20 is: 30

8. What is the use of source command in shell scripting?

Ans:

The source command in shell is used to execute commands from file in current shell environment instead of creating a new subshell

- Load environment variables from a file into the current shell.
- Run a script without starting a new shell, so any changes remain available after the script finishes.
- Useful for reusing functions or settings across multiple scripts or sessions

9. How can you debug a shell script? Give two methods.

Ans:

1.Run the script with `bash -x` to show each command and its arguments as it executes or add it at top of the script

Here each command is printed before execution making it easy to see where the script may fail

2. Use the `-n` option this runs the script to check for syntax error without executing it `bash -n`, bash checks for the errors like unmatched quotes or syntax mistakes

10. Write a bash script to create and delete a file.

```
#!/bin/bash
```

```
echo "Creating file testfile.txt..."
```

```
touch testfile.txt
```

```
if [ -f testfile.txt ]; then
```

```
    echo "File created successfully."
```

```
else
    echo "Failed to create file."
fi
echo "Deleting file testfile.txt..."
rm testfile.txt
if [ ! -f testfile.txt ]; then
    echo "File deleted successfully."
else
    echo "Failed to delete file."
fi
```

Output:

```
Creating file testfile.txt...
File created successfully.
Deleting file testfile.txt...
File deleted successfully.
```